

## Phase 4: Team Reflection (For Slot 2)

**Project Title:** Automated EOT Crane

**Category:** D

**Team Members:** H066,H067,H090,H094

**Q1: What was the single hardest technical challenge you faced in deriving the mathematical model? How did you resolve it?**

The hardest challenge was figuring out how to mathematically link the horizontal motion of the crane bridge to the swinging motion of the payload. Initially, we treated them as separate systems. We resolved this by defining the absolute spatial position of the payload relative to the moving bridge, and taking the second time derivative (acceleration) using the product rule. This revealed that the bridge's acceleration acts directly as a driving force on the pendulum.

**Q2: Which Course Outcome (CO1-CO5) was most challenging for your team? Why?**

CO4 (Apply the constitutive laws to analyse the dynamic systems of particles and rigid bodies) was the most challenging. While calculating pure kinematics (CO1) was straightforward, applying Newton's Second Law to a coupled system where the cable tension acts as both a vertical support and a horizontal restoring force required careful sign convention and a solid understanding of the small-angle approximation.

**Q3: What would you do differently if you started the project over?**

If we started over, we would begin by modelling a purely 2D version of the crane (just the trolley and payload) before attempting the full 3D spatial model. Jumping straight into 5 Degrees of Freedom made the initial coordinate mapping and velocity derivatives overwhelmingly complex.

**Q4: What did you learn from this project that a regular textbook problem could NOT teach you?**

Textbook problems usually provide a static formula for the natural frequency of a pendulum. This project taught us how to model the *source* of the disturbance. We learned that hoisting a load upwards (changing the cable length  $L$  dynamically) introduces a Coriolis acceleration term ( $2 * \dot{L} * \dot{\theta}$ ) that actively changes the swing behaviour, which is a critical reality in industrial crane operation.

**Q5: How did you divide work among team members? Did everyone understand the full derivation?**

H090 and H066 focused heavily on the Phase 1 Free Body Diagrams and the Newton-Euler derivations. H067 and H094 took the lead on translating those final EOMs into the Python `solve_ivp` simulation and conducting the energy validation checks. We held a team review session where the derivation team walked the coding team through the math step-by-step.

**Q6: What AI tools did you use? For what specific purpose? What did you NOT use AI for?**

We used the AI tool Gemini. We used it to help format and verify our step-by-step mathematical derivation of the Equations of Motion, and to translate our final equations into functional Python syntax using the `scipy.integrate.solve_ivp` library to generate our validation plots. We did NOT use AI to draw our Free Body Diagrams, nor did we submit any derivation steps or code that we could not physically explain or defend ourselves.